Dataset Documentation

Safety-Jacket Follower Robot | Jamugisa Peter Paul and Ayebare Moses | 30 October 2026

The full dataset is about 890 MB (about 38,000 image and label files), so it is documented here and not included in the submission. The training configuration is in `6_Source_Code/training_config`.

1. Target class and format

One class: `0: safety_jacket`, a hi-vis vest with reflective bands, worn or hanging. Labels are YOLO (You Only Look Once) text files with one line per box: class, x-centre, y-centre, width and height, each scaled to 0–1. An image with no vest has an empty label file, which makes it a negative example.

2. Training and validation sets (version 2, used by the final model)

Source	Images	Labels and notes
Robot-camera frames, session 1 (328 frames, used twice)	656	Grounding DINO plus colour filter plus human check; repeated to up-weight the deployment view
Original phone photos of one jacket	284	Original annotations
Public vest photos	1,500	Human labels from the public datasets below
Public photos of people without vests	700	Empty labels (negatives)
Total training images	**3,140**	
Validation images	**115**	HB1204 validation split (public, human-labelled)

3. Independent test sets (never used for training or model selection)

Test set	Images	Contents and labels
Robot camera, held-out time block (session 1, frames 230–330)	99	94 with a jacket, 5 without; Grounding DINO plus human check, independent of all YOLO models
Public set A (HB1204 test split)	242	186 with vests, 56 people without; human labels
Public set B (Vincent test and validation splits)	208	112 with vests, 96 people without; human labels, some vests unlabelled

4. Sources and licences

- **Our own data:** frames recorded on the Hiwonder MasterPi arm camera in the lab, plus our earlier phone photos. People in the frames are project members.
- **HB1204/PPE_Detection**, Hugging Face, CC BY 4.0 (Creative Commons Attribution 4.0): `https://huggingface.co/datasets/HB1204/PPE_Detection`. Original classes used: Safety Vest (10), NO-Safety Vest (7).
- **VincentGOURBIN/ppe-detection**, Hugging Face, CC BY 4.0: `https://huggingface.co/datasets/VincentGOURBIN/ppe-detection`. Original classes used: vest (4), no-vest (2).

5. How the robot frames were labelled

1. Session 1 gave 506 frames; 57 corrupt frames were removed, leaving 449 usable frames.
2. Grounding DINO (`IDEA-Research/grounding-dino-base`) proposed boxes from a text prompt.
3. A hi-vis filter in HSV (Hue, Saturation, Value) kept boxes with at least 8% of pixels in the range (30, 45, 80) to (70, 255, 255). When boxes were nested, the inner box was kept.
4. Results were checked by eye using contact sheets; two ambiguous test frames (00244, 00250) were removed.

6. Splits and leakage control

Neighbouring video frames are nearly identical, so a random split would leak near-copies into the test set. We used **time-block splits**: the robot test frames are a continuous block not used in training. An early evaluation used labels produced by the original model, which is circular, so we replaced it with these independent test sets.

7. Known limitations

- Robot frames come from one room, one person and one day, so a test in a new location is still needed.
- The robot test set has only 5 frames without a jacket, so its false-alarm figure is weak evidence.
- Public set B has some unlabelled vests, which can lower its measured precision.

8. Reproducing the dataset

`scripts/dino_label.py` labels frames, `scripts/get_public_test.py` downloads the public data, `scripts/prepare_dataset_v2.py` builds the training set, and `scripts/train_v2.py` trains the model. `scripts/evaluate_all.py` produces the results in `RESULTS.md`.